On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues.

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1) Consider the matrix $A=\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 3 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & \beta \end{bmatrix}$ Some of the elimination matrices are given below.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & \lambda \end{bmatrix}, \begin{bmatrix} \lambda & 0 & 0 & 0 \\ -\beta & 1 & 0 & 0 \\ m & 0 & \lambda & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -11 & 1 \end{bmatrix}$$

a) Find the unknown entries α, γ, β, k and m? (5)

b) Find the eliminated upper triangle matrix U? (5)

c) Determine the remaining elimination matrices. (10)

2) A geometry in 'k' dimensional space is given by x=5y expression, where 'x' and 'y' represent the axis. The instersection with another geometry gives a volume.

a) Give the minumum dimension of the space with your valuable comments. (10)

I think minumum dimension of the space is R^4. Because if intersection with another geometry gives a volume it means it is at least 3-D. But because of it is a intersection it is 4-D space.

b) Propose an expression for the unknown geometry to have a volume in intersection (10)

$x+4y+3z=8$

c) I proposed a geometry $x+4y-2z=6$. Find all the vectors in the intersection. Do these vectors form a vector space? The intersection has how many dimensions inside how many dimensional space. (15)

Forming vector space

- summable
- scalable

These vectors forms ~~a~~ vector space because both summable and scalable

2-D inside R^3

3) Suppose u is a unit vector in $\mathbb{R}^n$. This problem is about the n by n symmetric matrix $H = I - 2uu^T$

a) Show directly that $H^2 = I$. (5)

$$H = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 2 \cdot \begin{bmatrix} a \\ \sqrt{1-a^2} \end{bmatrix} \cdot \begin{bmatrix} a & \sqrt{1-a^2} \end{bmatrix}, \quad H^2 = \begin{bmatrix} 1-2a^2 & -2a\sqrt{1-a^2} \\ -2a\sqrt{1-a^2} & 1-2a^2 \end{bmatrix} \cdot \begin{bmatrix} x & y \\ y & x \end{bmatrix} = \begin{bmatrix} x^2+y^2 & xy+yx \\ xy+yx & y^2+y^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} a^2 & a\sqrt{1-a^2} \\ a\sqrt{1-a^2} & 1-a^2 \end{bmatrix}$$

$x^2 = 1-4a^2+4a^4$, $y^2 = 4a^2(1-a^2)$ $\Rightarrow 1 / (1-2a^2)(-2a)\sqrt{1-a^2} = 0$

b) Show that H is an orthonormal matrix. (5)

H is an orthonal matrix because $H^T \cdot H = I$. Thats orthonormal formula. $(q_i^T \cdot q_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases})$

$H^T = H$ is the same. As I show in part A $H^2 = H^T \cdot H = I$

c) One eigen vector of H is u itself. Find the corresponding eigen value. (5)

$\begin{bmatrix} a & b \\ b & c \end{bmatrix} \cdot \begin{bmatrix} u \\ u \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow \lambda = 0$. Because $[H] \cdot [u] = 0$

d) Is H diagonalizable? why or why not? (5)

Yes. Because this is symmetric so we can digonalize this.

4) Two tanks are connected, first tank has $v(0) = 10$ liters liquid and the second tank has $w(0) = 40$ liters of water. The liquid diffusion between the tanks are proportional to the difference v.w.

a) which one of the following differential equation set express the process? (v_k: liquid diffused the 2nd tank, w_k: the water diffused to the second tank.) (10)

$$\begin{bmatrix} v_{k+1} \\ w_{k+1} \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} v_k \\ w_k \end{bmatrix} \qquad \frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix} \qquad \frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix}$$

b) Determine percentages of the mixture in tank1 at time $t = 1$ sec. (10)

$|\lambda I - A| = 0 \quad \begin{vmatrix} \lambda+2 & -1 \\ -1 & \lambda+2 \end{vmatrix} = 0 \quad (\lambda+2)^2 - 1 = 0$

$\lambda_1 = -1 \rightarrow \begin{vmatrix} 1 & -1 \\ -1 & 1 \end{vmatrix} \cdot x = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\lambda_2 = -3 \quad \begin{vmatrix} -1 & -1 \\ -1 & -1 \end{vmatrix} \cdot x = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$\begin{bmatrix} v_k \\ w_k \end{bmatrix} = c_1 \cdot (-1)^n \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \cdot (-3)^n \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ for $n = 0 \rightarrow c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 10 \\ 30 \end{bmatrix}$

$c_1 + c_2 = 10$, $c_1 - c_2 = 30$, $c_1 = 15$, $c_2 = -15$

$= 15 \cdot (-1)^1 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 15 \cdot (-3)^1 \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 50 \\ -50 \end{bmatrix}$

c) Find how much liters of liquids are exist in each tank in steady state?

$15 \cdot (-1)^{\infty} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 15 (-3)^{\infty} \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix} = ?$

25 liters in each tank.